DevOps 2TIN

Chapter 4

Pipelines

DE HOGESCHOOL MET HET NETWERK

Elfde-Liniestraat 24, 3500 Hasselt, www.pxl.be

Pipelines

Wat is het
stappen
Automatisch gebouwd
Reflectie

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Wat is een Pipeline?

Definitie “deployment pipeline”:

(first defined by Jez Humble and David Farley in their book *Continuous Delivery: Reliable Software Releases Through Build, Test, and Deployment Automation*)

It ensures that all code checked in to version control is automatically built and tested in a production-like environment.

Sleutelwoorden:

- Alle Code
- Versiebeheer
- **Automatisch gebouwd**
- Automatisch getest
- Productie-waardige omgeving

THE FIRST WAY

CREATE A DELIVERY SYSTEM

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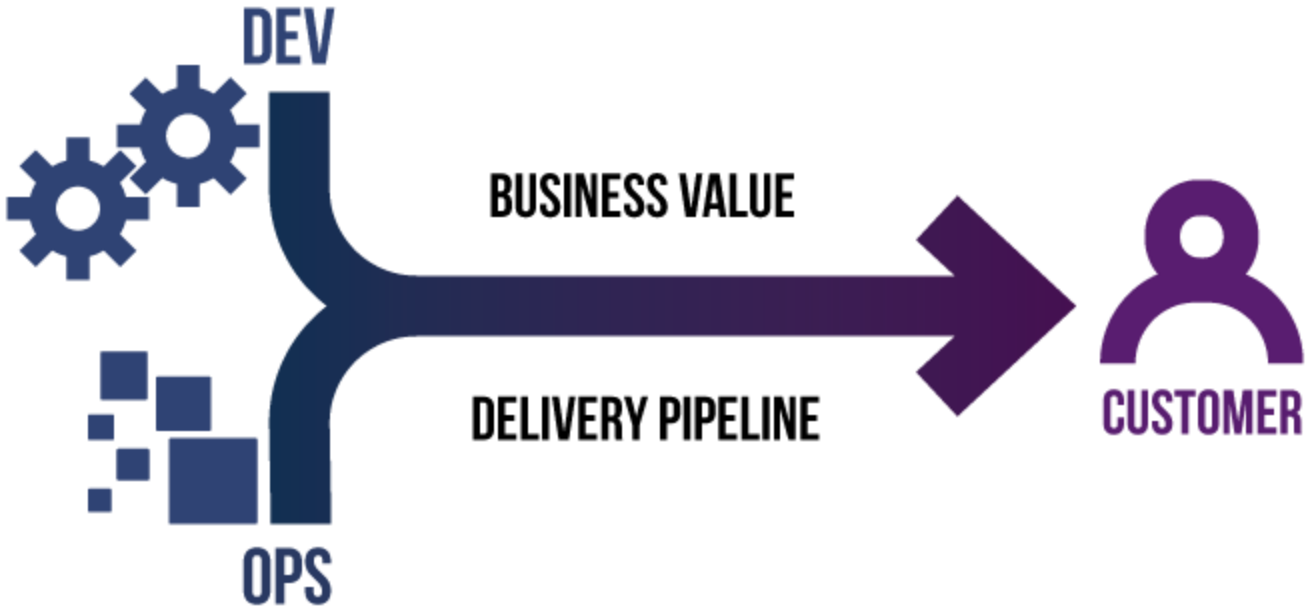
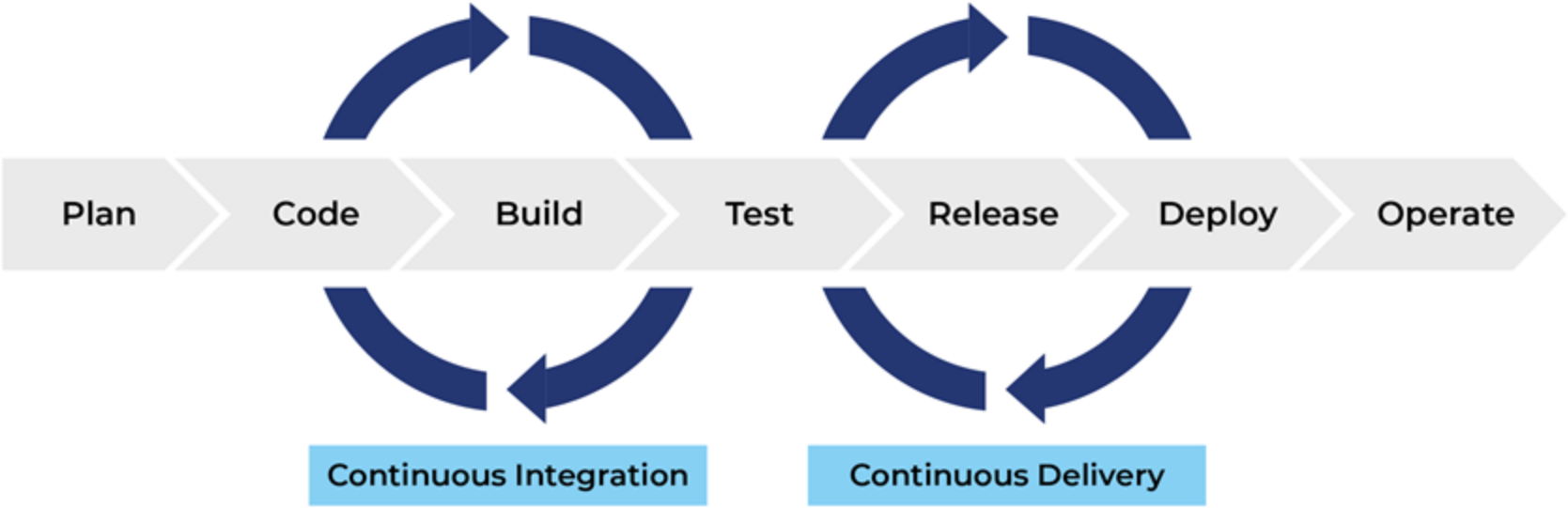


Developers practicing ***continuous integration*** merge their changes back to the main branch as often as possible.

CI/CD



The developer's changes are validated by creating a build and running **automated tests** against the build. By doing so, you avoid the integration hell that usually happens when people wait for release day to merge their changes into the release branch

Continuous integration

Code binnenhalen
Dependencies installeren
Builden / Packagen van code
Artifacts / packets beheer

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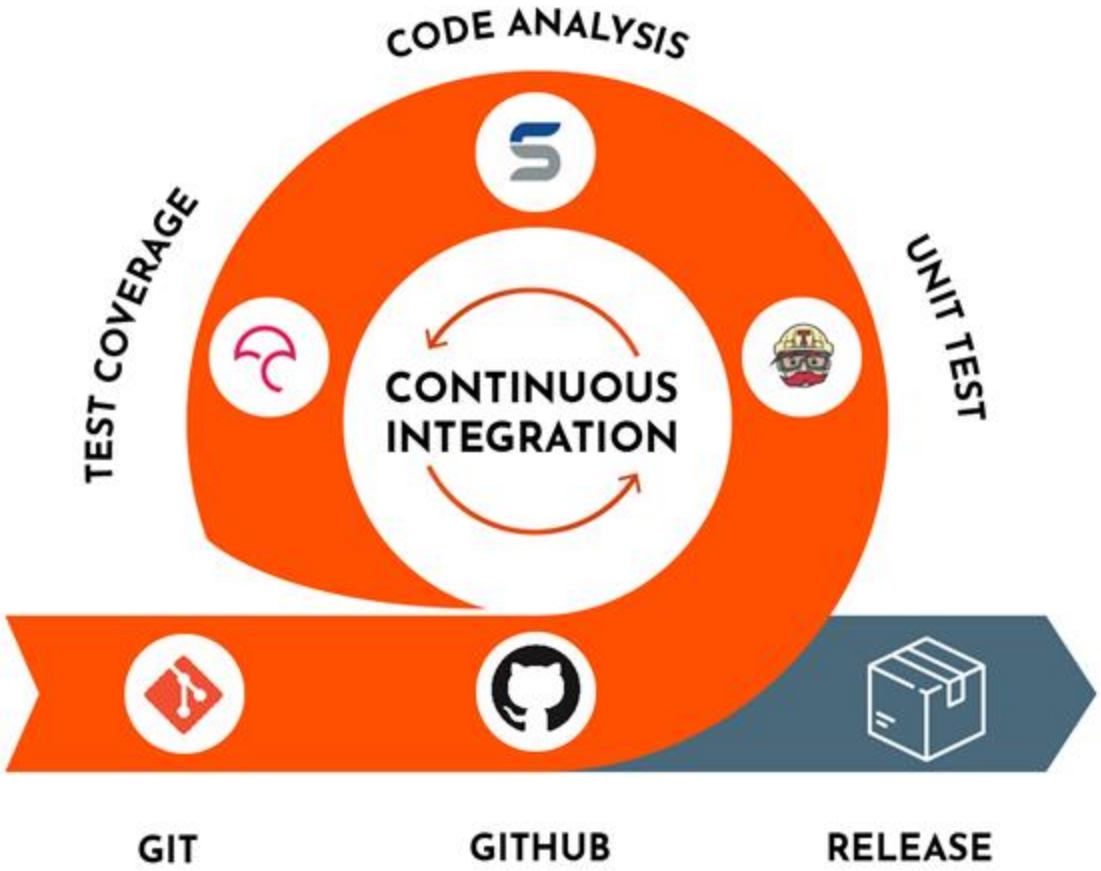
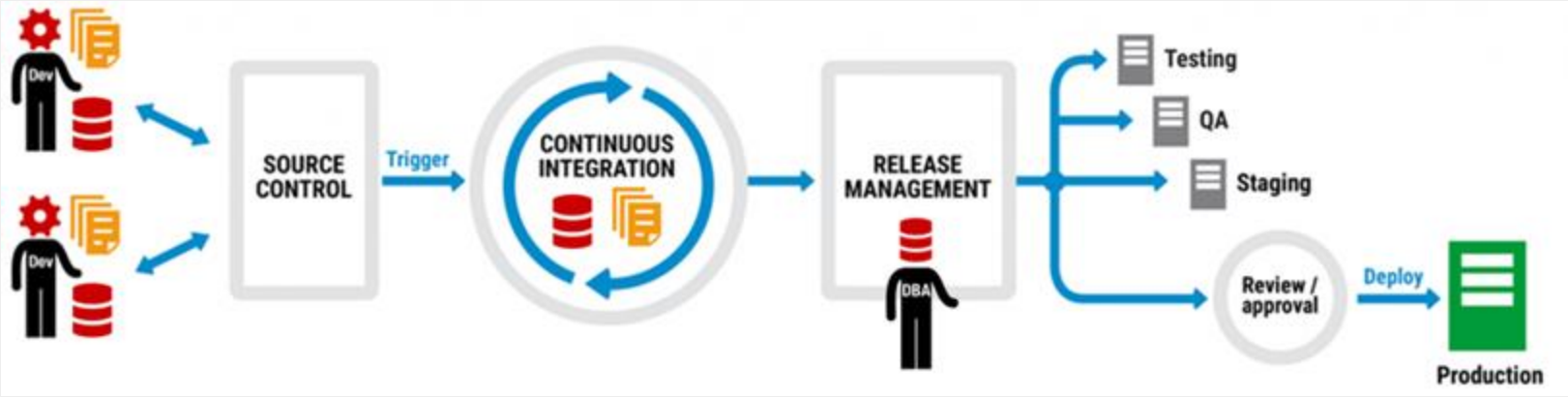

CODE ANALYSIS
TEST COVERAGE
UNIT TEST
CONTINUOUS INTEGRATION
GIT
GITHUB
RELEASE

Fetching code

- Binnenhalen van code vanuit source control
  - -> Brengt deze in de pipeline



Code	Dependencies	(unit)test	build	archive artifact	Deploy to QA	Deploy to production
CI					CD	

Installing dependencies

Bundelen van dependencies in de omgeving

- Afhankelijk van de programmeertaal / omgeving
  - PHP -> Composer install
  - Javascript -> npm install
  - ...

Code	Dependencies	(unit)test	build	archive artifact	Deploy to QA	Deploy to production
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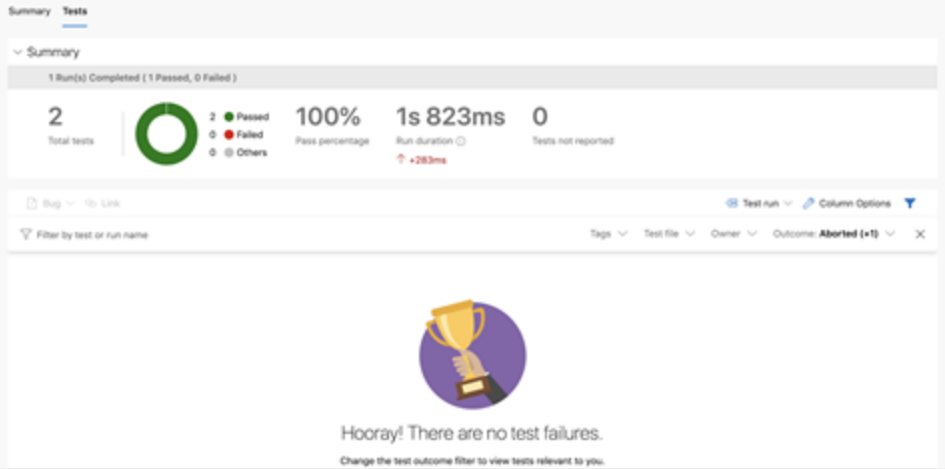
Test

- automatisatie van testen m.b.t. code
  - Code quality
  - Linting
  - Unittesten
  - …
- Geen QA omgeving nodig



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Build

- Automatiseren van het compileren & builden applicatie
  - Afhankelijk van de programmeertaal / omgeving
    - java -> mvn/ant -> jar
    - net core -> dotnet publish -> exe
    -

Code	Dependencies	(unit)test	build	archive artifact	Deploy to QA	Deploy to production
CI					CD	

Archiving artifacts / packaging

- Beheer van pakketten
  - Centraal? Verschillende versies/Per Build?
  - Build once, deploy anywhere

App	Development	Quality Assurance	User Acceptance	Production
SCM	deployment succeeded	deployment succeeded	deployment failed	deployment succeeded
App	UPack 1.0.7	UPack 1.0.7	UPack 1.0.6	UPack 1.0.4
Sources	UPack 1.0.7	UPack 1.0.7	UPack 1.0.6	UPack 1.0.4

Code	Dependencies	(unit)test	build	archive artifact	Deploy to QA	Deploy to production
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Continuous Delivery / Deployment

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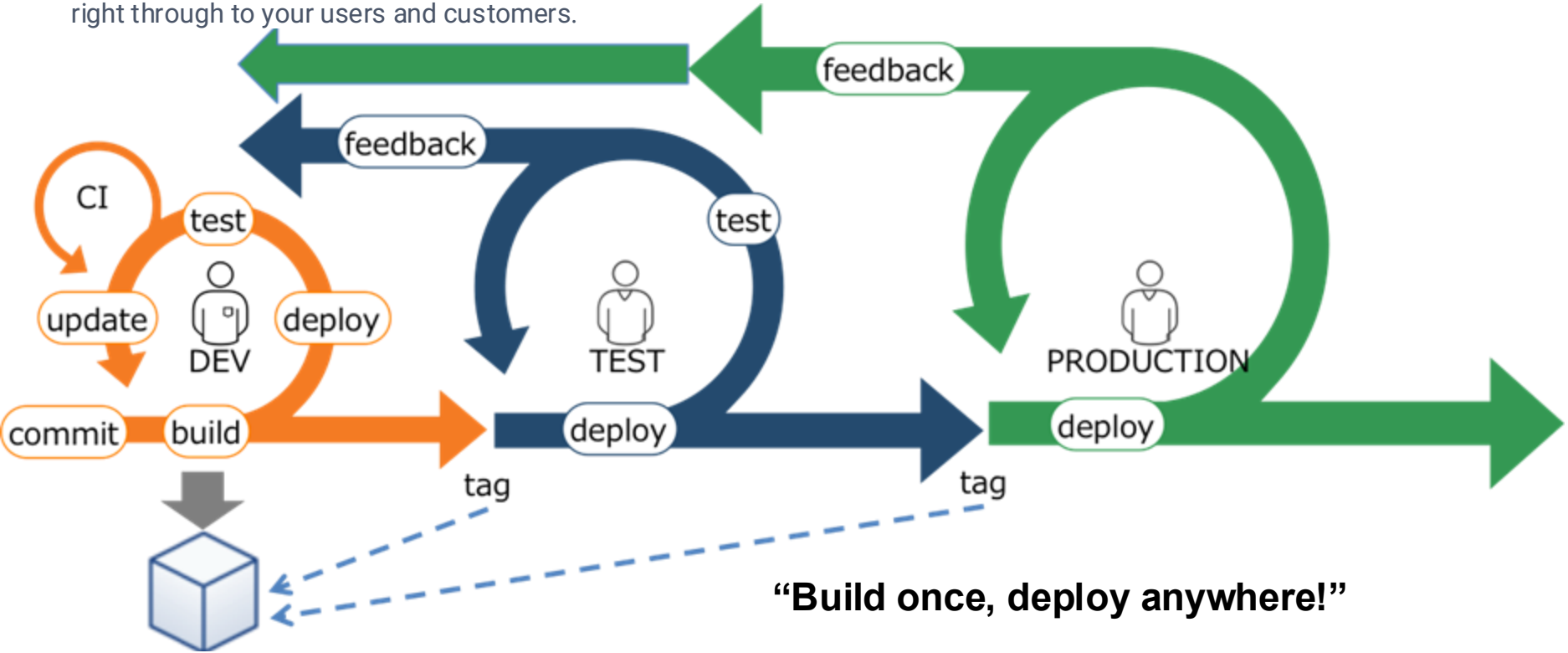
CI/CD

A ***continuous delivery (CD) pipeline*** is an automated expression of your process for getting software from version control right through to your users and customers.



“Build once, deploy anywhere!”

Deploying Software

Wat is er nodig?

- gecompileerde code
- omgeving specifieke parameters
- dependencies
- wat nog?

Code	Dependencies	(unit)test	build	archive artifact	Deploy to QA	Deploy to production
CI					CD	

Deploying Software

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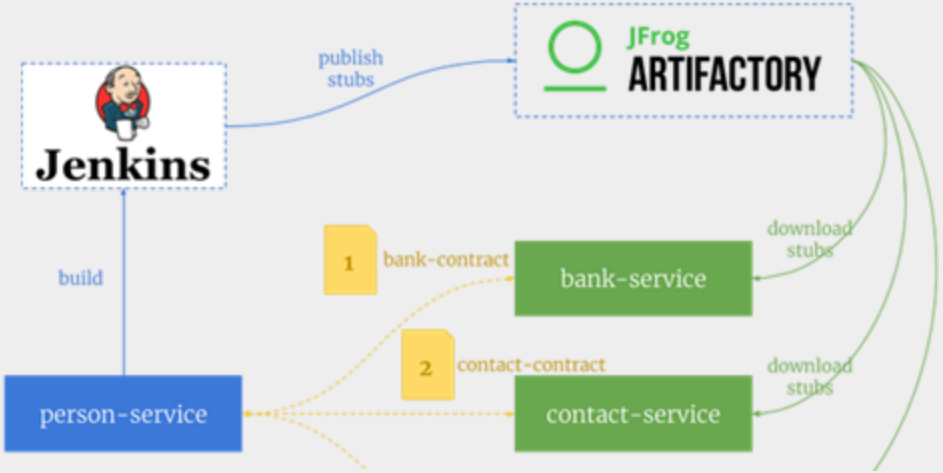
Automatiseren van de deploy

- Vertrekken vanuit de artifact vanuit onze CI pipeline
  - Centrale repository?



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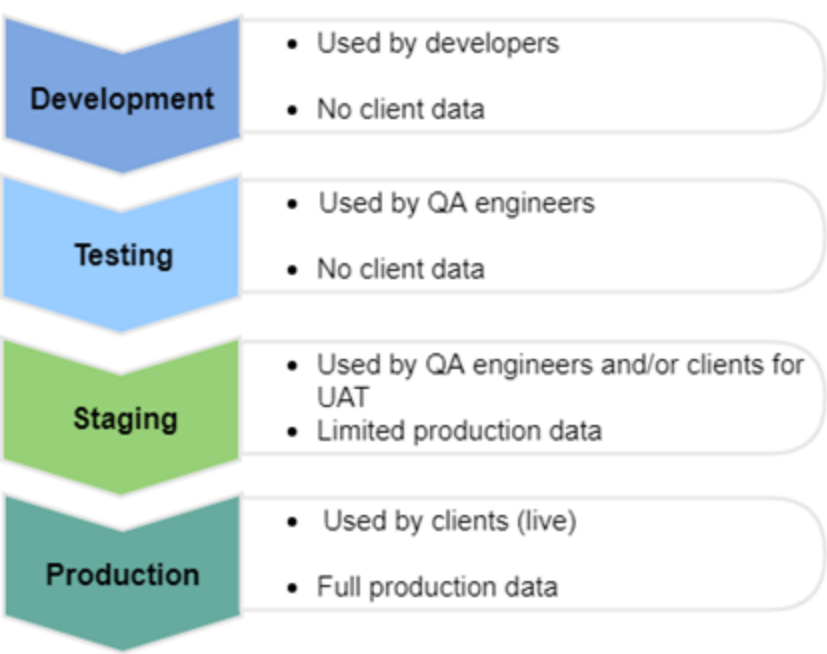
Deploying Software

Automatiseren van de deploy

- Omgeving in orde maken
  - Dependencies
  - omgeving specifieke parameters, credentials, secrets, ...



Code	Dependencies	(unit)test	build	archive artifact	Deploy to QA	Deploy to production
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CI / CD Pipeline

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- Structuur ligt niet vast
  - Elk bedrijf / omgeving heeft zijn eigen accenten

Code	Dependencies	(unit)test	build	archive artrifact	Deploy to QA	Deploy to production
CI					CD	

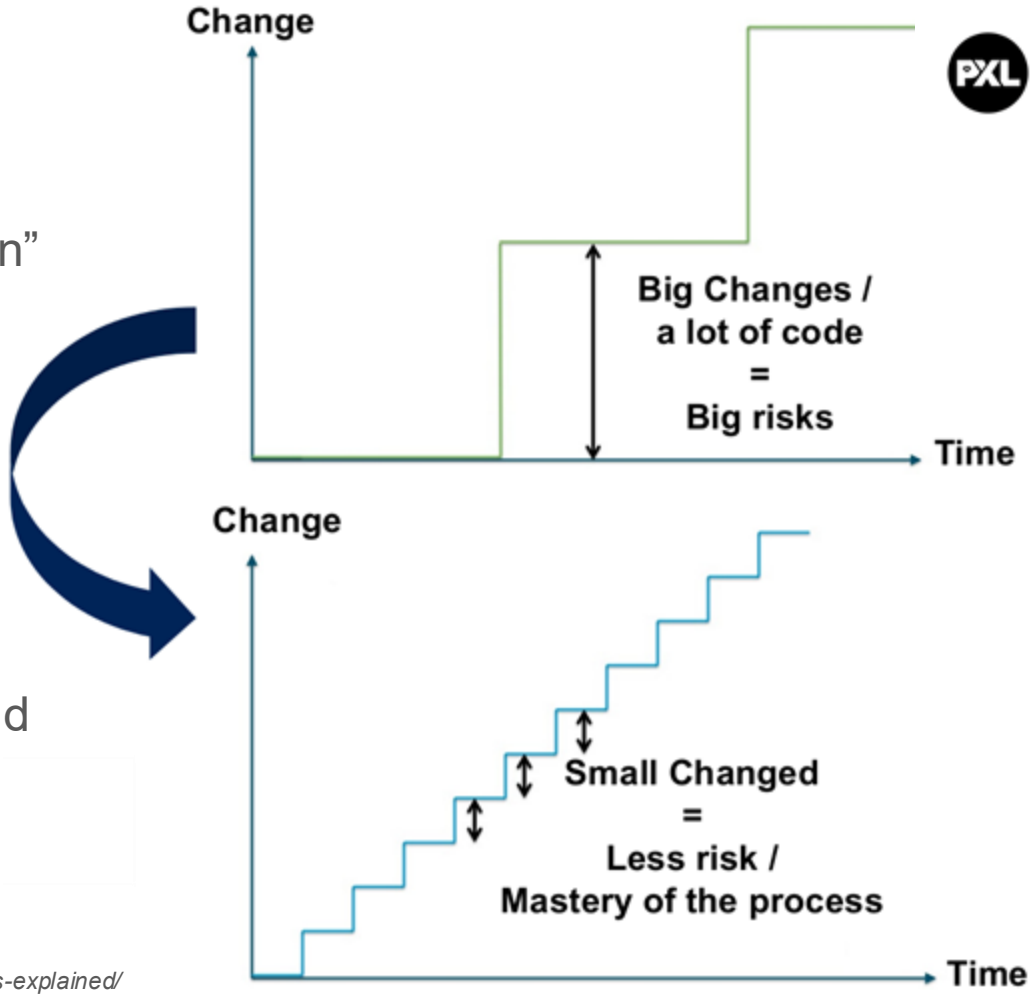
Maar Waarom?

“If it hurts, do it more often”

Vergeet apps!

Welkom:

- (Micro-)Services
- APIs
- Front-end & Back-end
- MDM



Source: https://www.devops.ch/2017/05/10/devops-explained/

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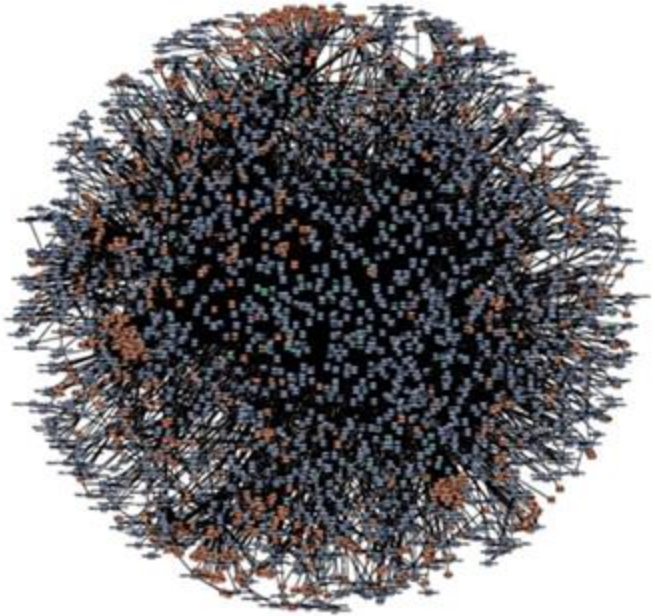

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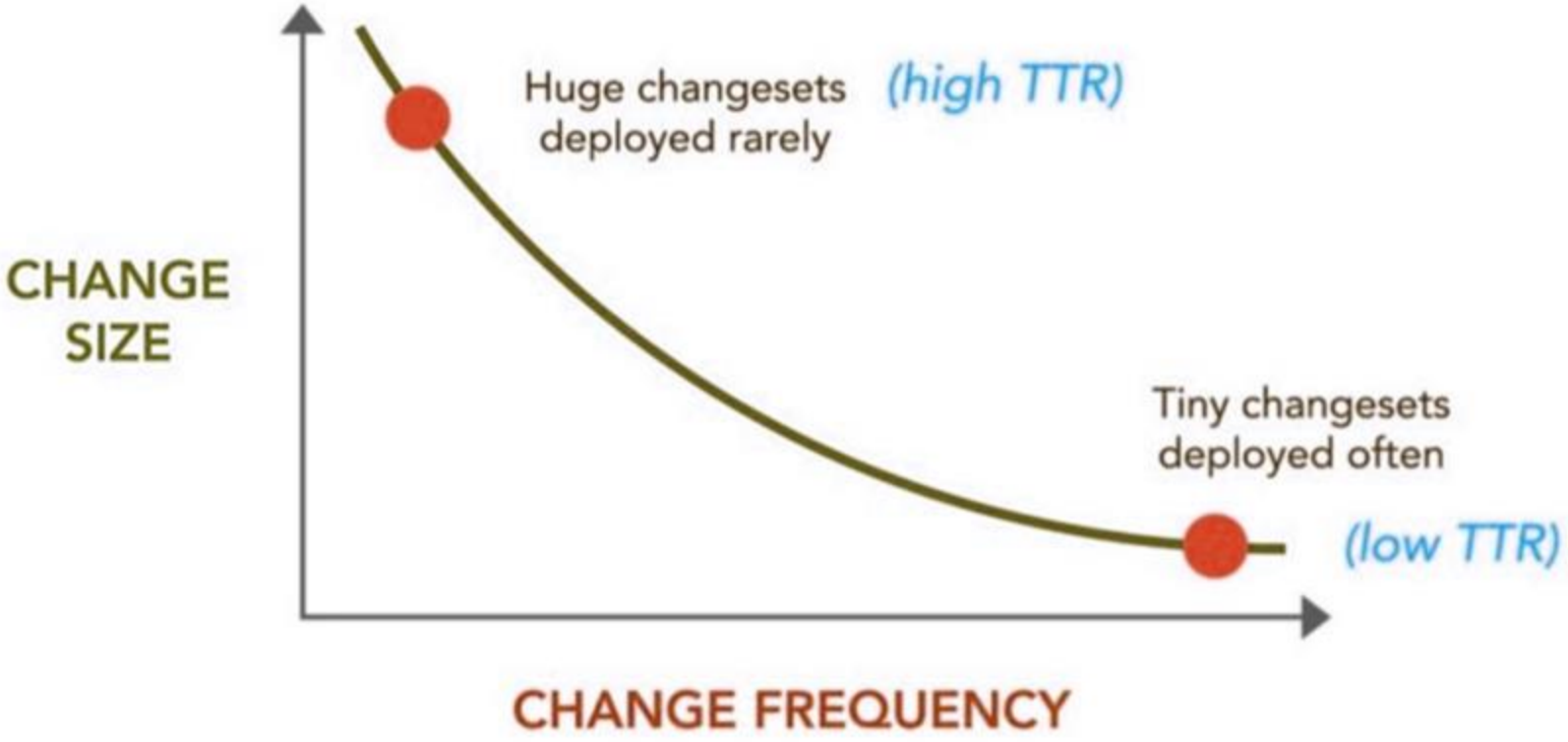
Maar waarom?



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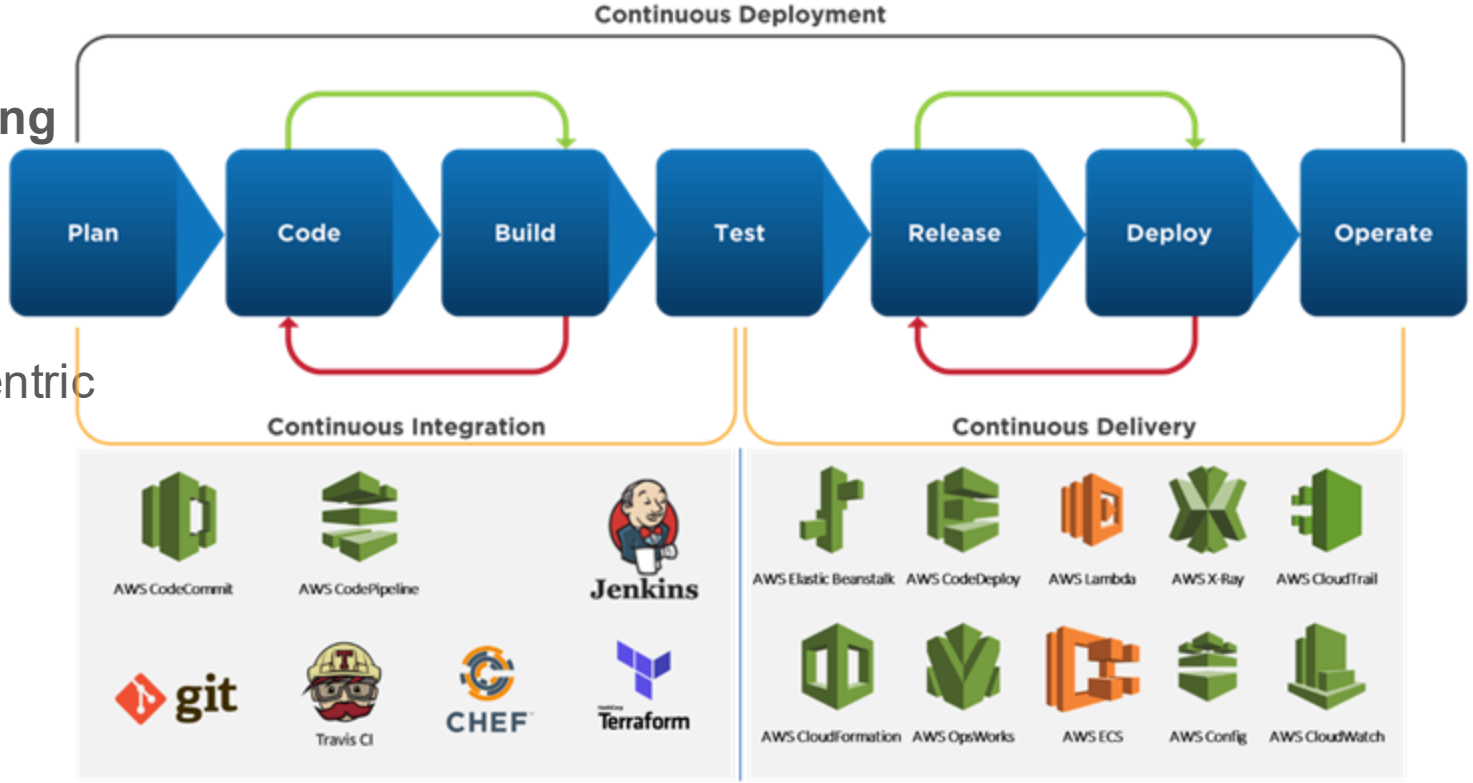
Maar hoe?

Pipeline tooling

- Oldskool
- Vendor Agnostic

Pipeline Tooling

- Newskool
- Vendor-centric



Dus...Jenkins

- Free & Open Source
- Automation Server
- parts of software development:
  - Building
  - Testing
  - Deploying

Jenkins

Software

Jenkins is a free and open source automation server. It helps automate the parts of software development related to building, testing, and deploying, facilitating continuous integration and continuous delivery. It is a server-based system that runs in servlet containers such as Apache Tomcat. Wikipedia

License: MIT License

Initial release: 2 February 2011

Platform: Java 8, Java 11

Stable release: 2.240 / 8 June 2020; 2 months ago

Written in: Java

Original author: Kohsuke Kawaguchi

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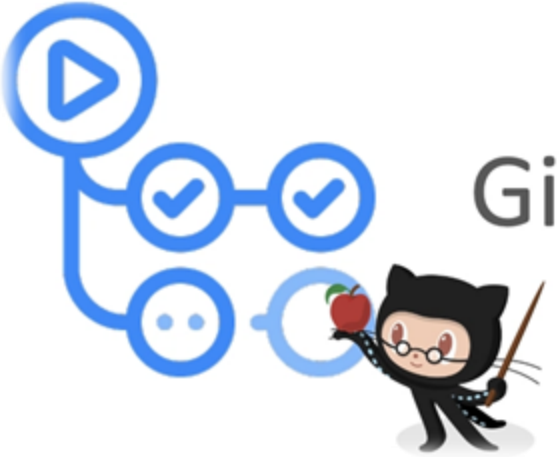
Feedback

Of toch niet?

- ~~Free & Open Source~~
  Semi
- Automation ~~Server~~
  Service
- parts of software development:
  - Building
  - Testing
  - Deploying

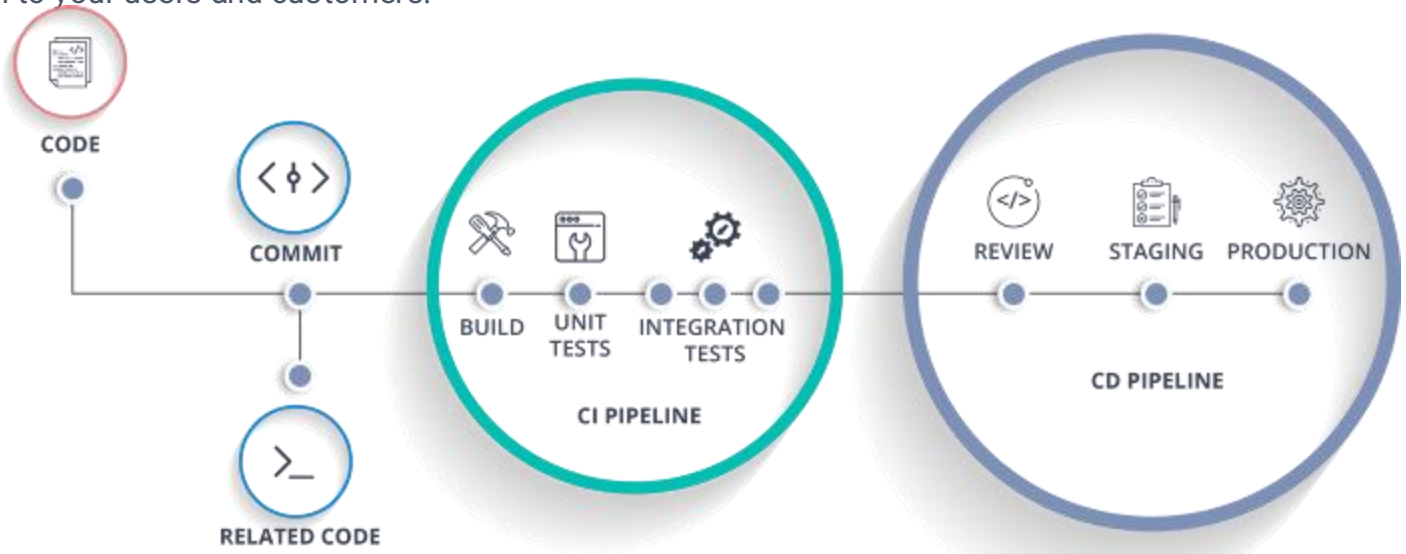


GitHub Actions

2018	4 June	Company	Microsoft announced it is acquiring GitHub.
	16 August	Product	GitHub Actions is launched[157]
2019	23 May	Acquisition	GitHub acquired dependabot.[158]
	18 September	Acquisition	GitHub acquired semmle.[159]
2020	16 March	Acquisition	GitHub announced that they are acquiring npm.[160]
2023	28 Jan	Usage	GitHub reports having ~100 million users, making it the largest host of source code in the world

Recap: CI/CD pipeline

A ***continuous delivery (CD) pipeline*** is an automated expression of your process for getting software from version control right through to your users and customers.



Every change to your software (committed in source control) goes through a complex process on its way to being released. This process involves building the software in a reliable and repeatable manner, as well as progressing the built software (called a "build") through multiple stages of testing and deployment.

Assignments

Lab 1 - Build requirements

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